

Property Support Assistant

A RAG-based customer support assistant built with React, ASP.NET Core, Azure OpenAI, and Azure services.

Project Overview

This project demonstrates a customer support chatbot for a property-search domain. It answers user questions from support documentation, shows source references, and returns a safe fallback for unsupported questions.

Key Features

- Chat-based support UI
- Example prompts for common issues
- Source-aware answers
- Fallback handling for unsupported questions
- Swagger API testing
- Azure monitoring and document storage

Tech Stack

Frontend: React, TypeScript

Backend: C#, ASP.NET Core Web API

Azure: Azure OpenAI, Blob Storage, Application Insights, App Service

Main API Endpoints

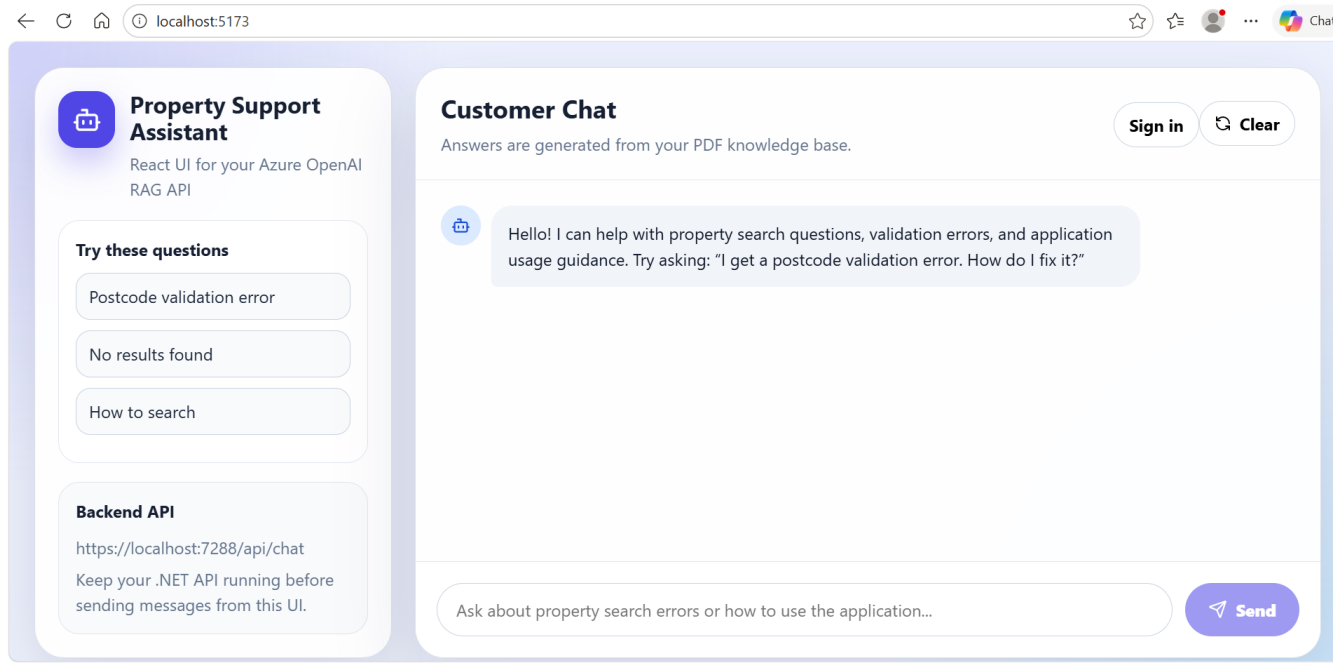
POST /api/chat

GET /api/health

GET /api/knowledge/search

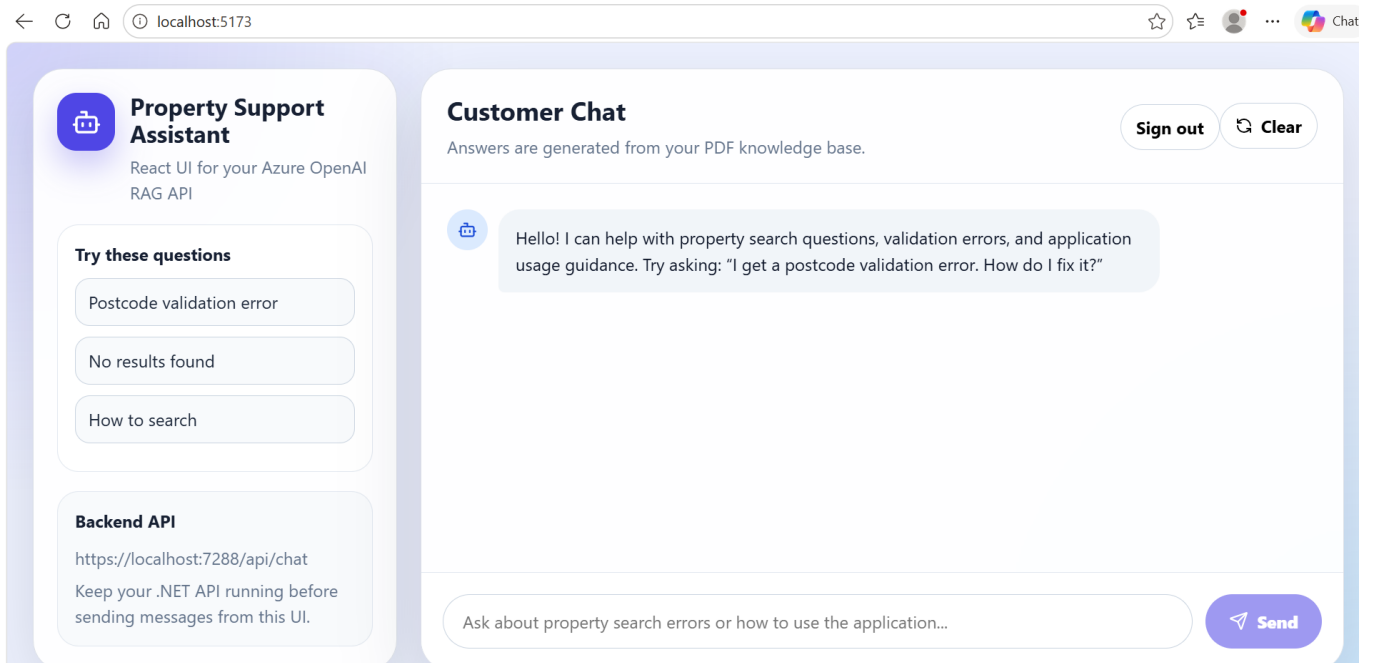
1. Home Screen

Initial chat UI with example questions and backend API information.



2. Signed-In View

Signed-in state of the application.



3. Supported Question Example

A domain-specific question returning a grounded answer.

The screenshot shows a web browser at localhost:5173 displaying the 'Property Support Assistant' interface. On the left, the 'Property Support Assistant' sidebar includes a React UI for Azure OpenAI RAG API, a 'Try these questions' section with buttons for 'Postcode validation error', 'No results found', and 'How to search', and a 'Backend API' section with the URL 'https://localhost:7288/api/chat' and instructions to keep the .NET API running. The main 'Customer Chat' area shows a user message: 'I get a validation error when searching by postcode. How do I fix it?'. The assistant's response explains the issue and provides five steps: 1. Ensure you enter a full UK postcode, e.g., SW1A 1AA. 2. Add a space before the last three characters if missing. 3. Remove any special characters, like commas or slashes. 4. Trim extra spaces at the start or end of the postcode. 5. If partial postcodes are not accepted, enter the complete postcode. It also advises contacting support if the issue persists. The chat input field contains the text 'Ask about property search errors or how to use the application...' and a 'Send' button.

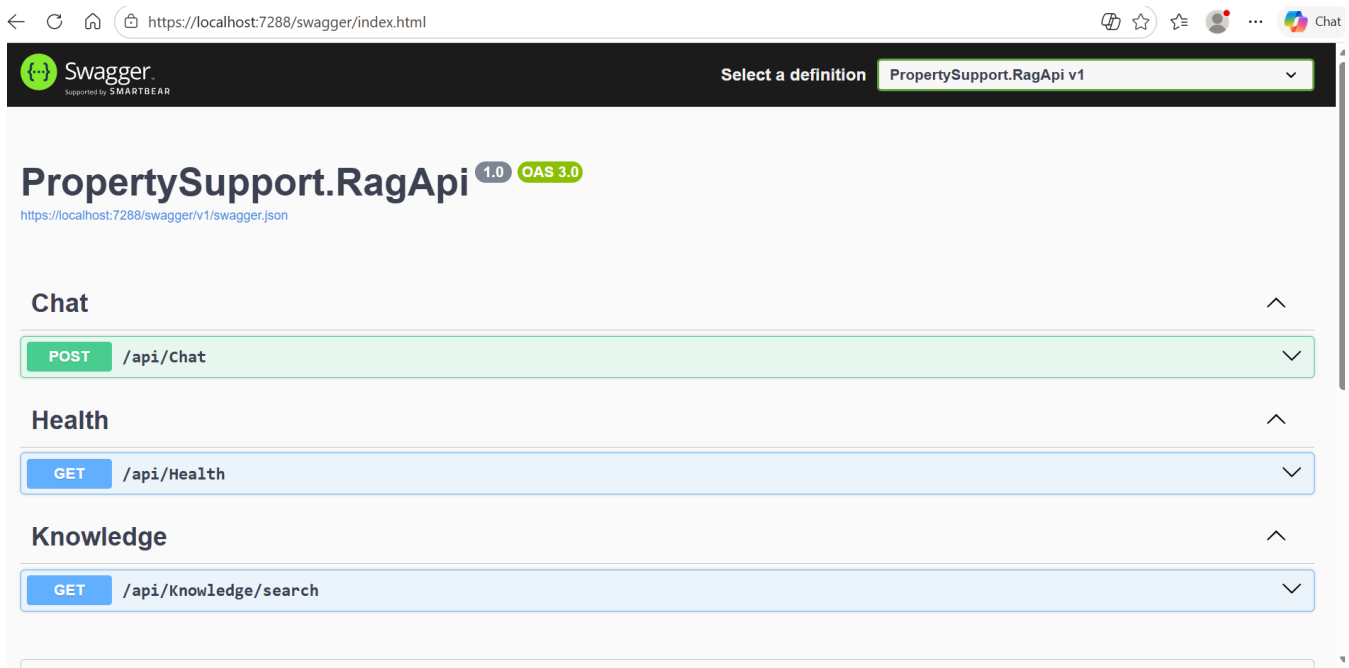
4. Unsupported Question Example

Fallback message for an out-of-scope question.

The screenshot shows the same web browser at localhost:5173, but the user has asked an out-of-scope question: 'how can I fix my phone'. The assistant's response is a fallback message: 'I could not find this in the support documentation. Please contact support.' Above and below this message, the 'Sources' section displays 'property_search_business_rules_demo.pdf' with scores of 7.03 and 1.92 respectively. The chat input field contains the text 'Ask about property search errors or how to use the application...' and a 'Send' button.

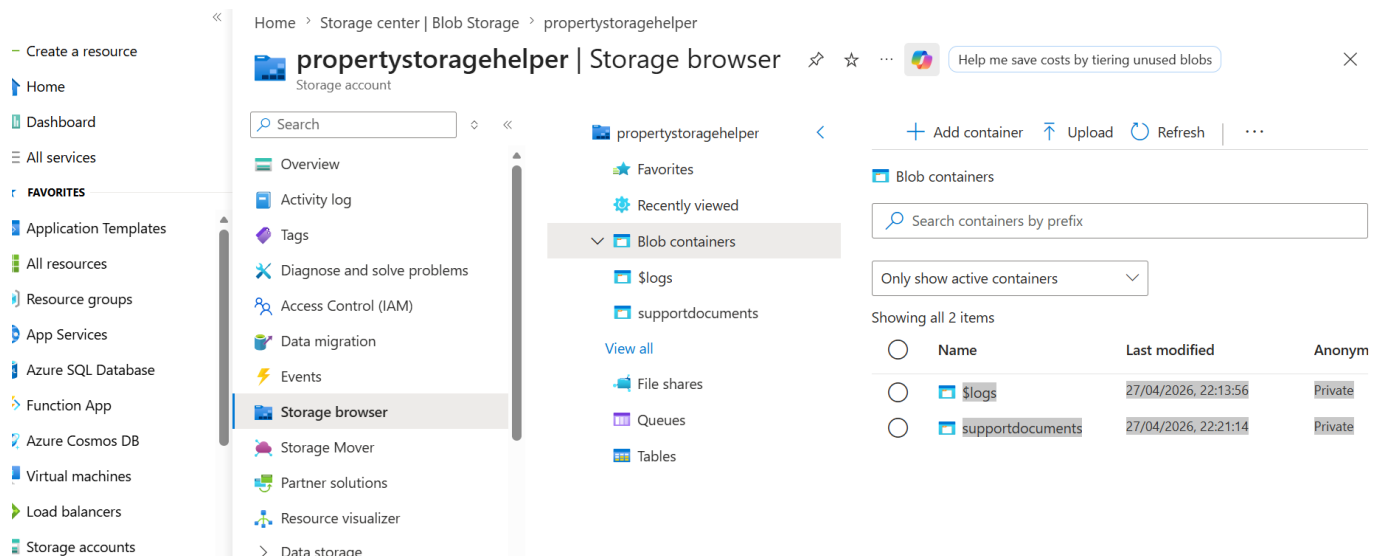
5. Swagger API

Backend endpoints exposed for testing.



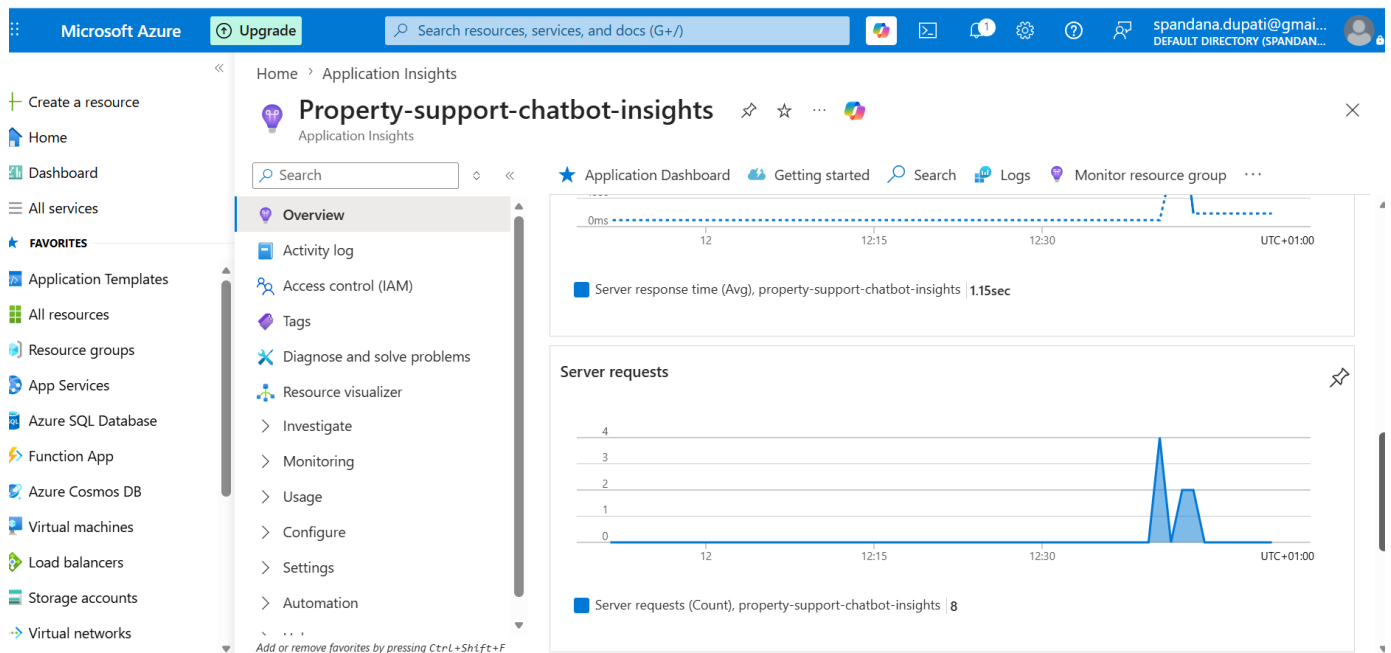
6. Azure Blob Storage

Support documents stored in Azure Blob Storage.



7. Application Insights

Monitoring dashboard showing request telemetry.



Outcome

This project shows an end-to-end AI support assistant combining a frontend UI, backend API, RAG workflow, Azure storage, observability, and safe handling of unsupported queries.